

# Competing Risks Survival Analysis: Mortgage Prepayment and Default

IR&C Assignment 3, Part E(i) — Spring 2026

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May 2026

## 1. Introduction

Mortgage prepayment (refinancing or sale) and default (foreclosure, short sale) are *competing risks*: the occurrence of one permanently prevents the other. Treating them as a single event — or analyzing each independently with the Kaplan–Meier estimator — biases every probability estimate and conflates two economically opposite processes. This report applies the correct competing-risks framework to 34 million Freddie Mac 30-year fixed-rate mortgages.

**The central finding:** four major covariates (FICO, LTV, delinquency, investment property) have statistically significant *opposite signs* for prepayment versus default. A combined model misrepresents both mechanisms.

## 2. Data

| Outcome                           | N                 | Share |
|-----------------------------------|-------------------|-------|
| Prepaid (ZBC = 01)                | 21,970,748        | 64.6% |
| Defaulted (ZBC ∈ {02,03,09})      | 532,563           | 1.6%  |
| Censored (still active)           | 11,510,158        | 33.8% |
| <b>Total (FRM-360, 1999–2023)</b> | <b>34,013,469</b> |       |

Covariate snapshots by event type confirm two distinct populations: prepaid loans have higher FICO (+44 vs defaulters), lower LTV (−8.6 pp), and lower origination rate (−85 bp). Cox and CIF models use a **5% stratified sample (≈1.7M loans)** drawn proportionally by vintage year.

## 3. Two Competing Causes: Exploratory Analysis

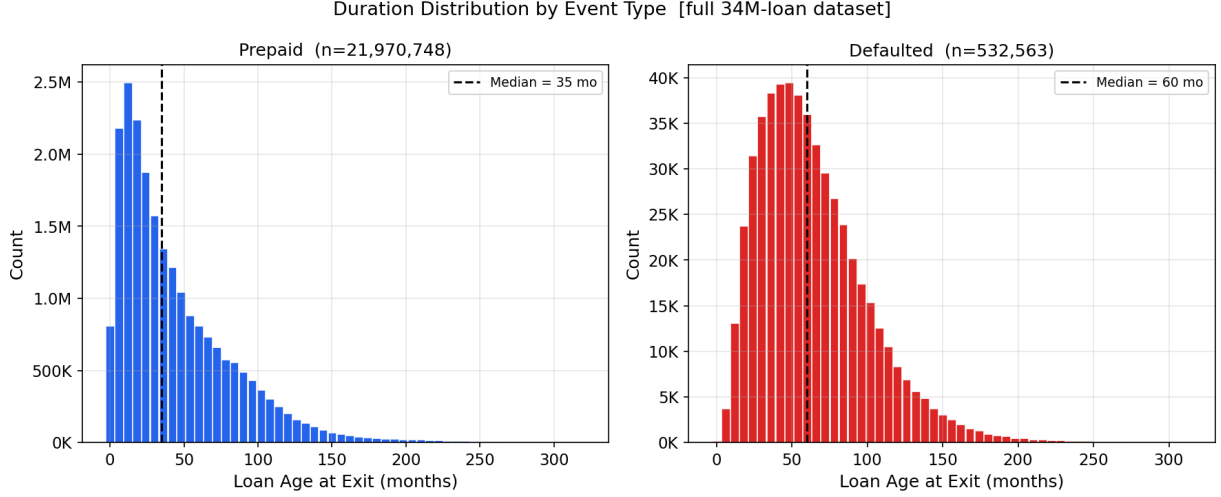
## 4. Aalen–Johansen CIF and the KM Bias

### 4.1 Why KM fails under competing risks

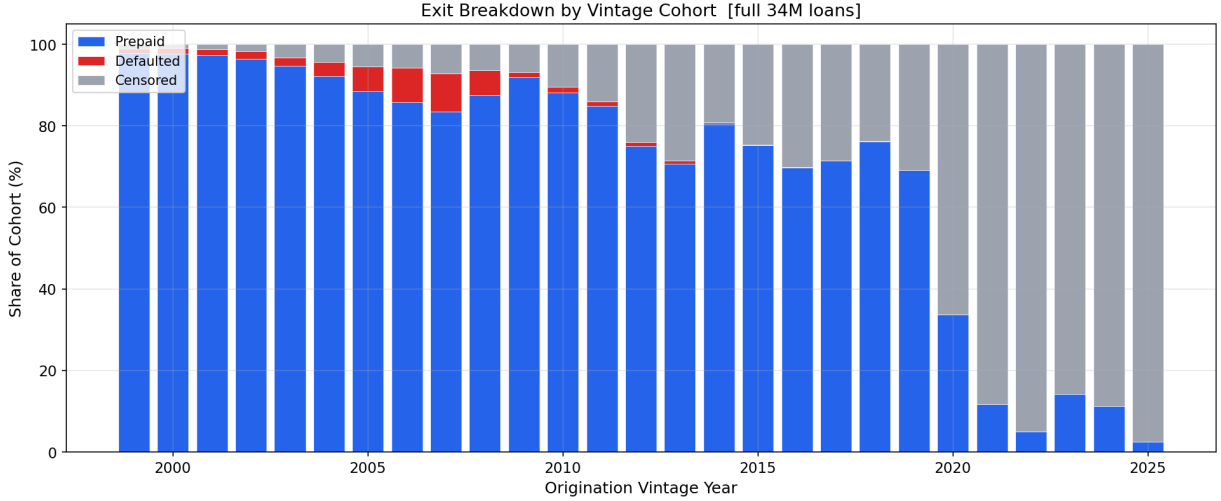
The Kaplan–Meier estimator treats competing exits as randomly censored, inflating each cause-specific probability:

$$\hat{F}_{\text{KM}}^{(k)}(t) = 1 - \prod_{t_j \leq t} \left( 1 - \frac{d_j^{(k)}}{n_j^{\text{KM}}} \right). \quad (1)$$

Defaulted loans — the *least likely to prepay* — are dropped from  $n_j^{\text{KM}}$ , so every KM increment exceeds the true value. The bias compounds:  $\hat{F}_{\text{AJ}}^{(\text{pre})}(t) \leq \hat{F}_{\text{KM}}^{(\text{pre})}(t)$  for all  $t$ .



**Figure 1:** Duration distributions by event type (full 34M dataset). Prepayment peaks at  $\sim 24$  months (rate-option exercise is front-loaded); default peaks at  $\sim 60$  months (gradual financial deterioration). The fundamentally different shapes imply different hazard functions — a single-baseline Cox model cannot represent both simultaneously.



**Figure 2:** Vintage cohort analysis. The 2007 cohort reaches  $\approx 8.5\%$  default —  $6\times$  the long-run average of  $\approx 1.4\%$  — driven by GFC underwriting. Post-2020 vintages show a large censored share: borrowers who originated at 3% face a strong rate lock-in disincentive to refinance.

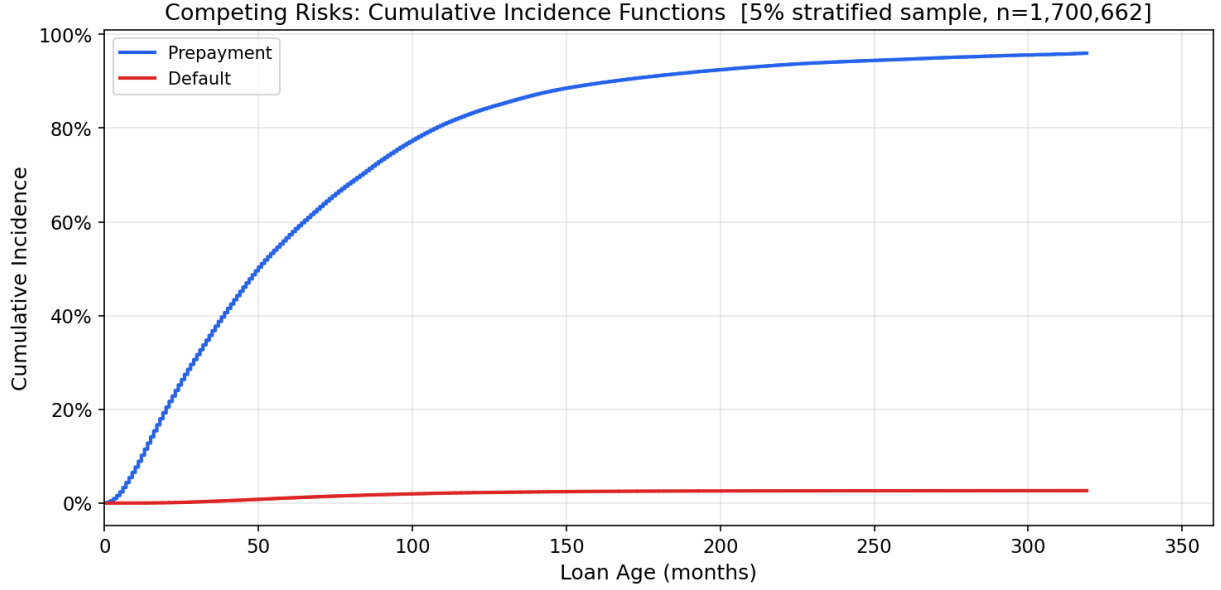
## 4.2 The Aalen–Johansen estimator

$$\hat{F}_k(t) = \sum_{t_j \leq t} \hat{S}(t_j^-) \cdot \frac{d_{kj}}{n_j} \quad (2)$$

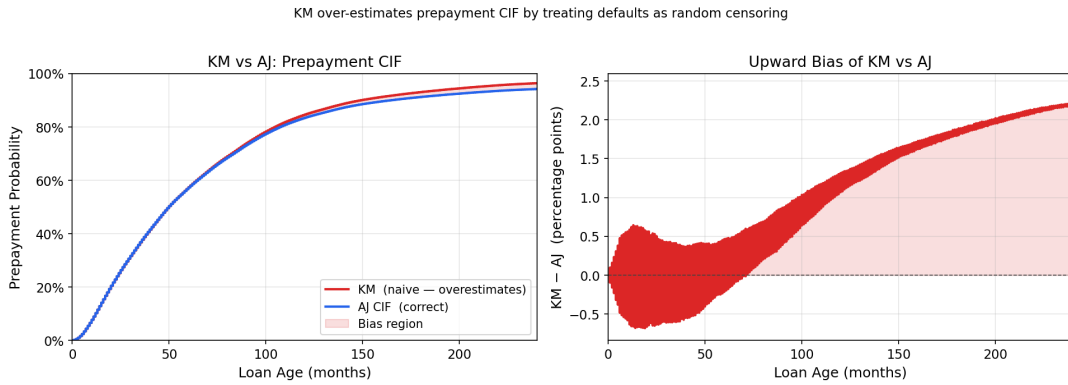
$\hat{S}(t_j^-)$  uses *all* exits to weight each increment, satisfying the partition identity  $\sum_k \hat{F}_k(t) + \hat{S}(t) = 1$  for all  $t$  — impossible under KM.

## 4.3 Results

| Horizon | $\hat{F}_{AJ}^{(\text{pre})}$ | $\hat{F}_{AJ}^{(\text{def})}$ | Still active | KM bias  |
|---------|-------------------------------|-------------------------------|--------------|----------|
| 5 yr    | 61.4%                         | 1.05%                         | 37.6%        | +0.45 pp |
| 10 yr   | 83.3%                         | 2.23%                         | 14.5%        | +1.31 pp |
| 20 yr   | 91.8%                         | 3.10%                         | 5.1%         | +2.23 pp |



**Figure 3:** Aalen–Johansen CIF. Prepayment dominates 37:1 at 10 years. The prepayment curve shows burnout: slope decelerates after month  $\approx 40$  as the most rate-sensitive borrowers exit. Default grows nearly linearly.



**Figure 4:** KM vs. AJ prepayment CIF. The gap widens monotonically as more defaulted loans accumulate and are mis-treated as random censoring. A 1 pp over-statement of the 10-year prepayment probability translates to  $\approx \$1\text{M}$  of mispriced duration in a  $\$100\text{M}$  MBS pool.

## 5. Risk Heterogeneity: Stratified CIFs

**FICO** separates default by  $6\times$ : default CIF ranges from  $\approx 1\%$  ( $\text{FICO} > 740$ ) to  $\approx 6\%$  ( $\text{FICO} < 680$ ) at 20 years. Prepayment CIFs converge across FICO buckets: FICO enables refinancing *access*, not refinancing *desire*.

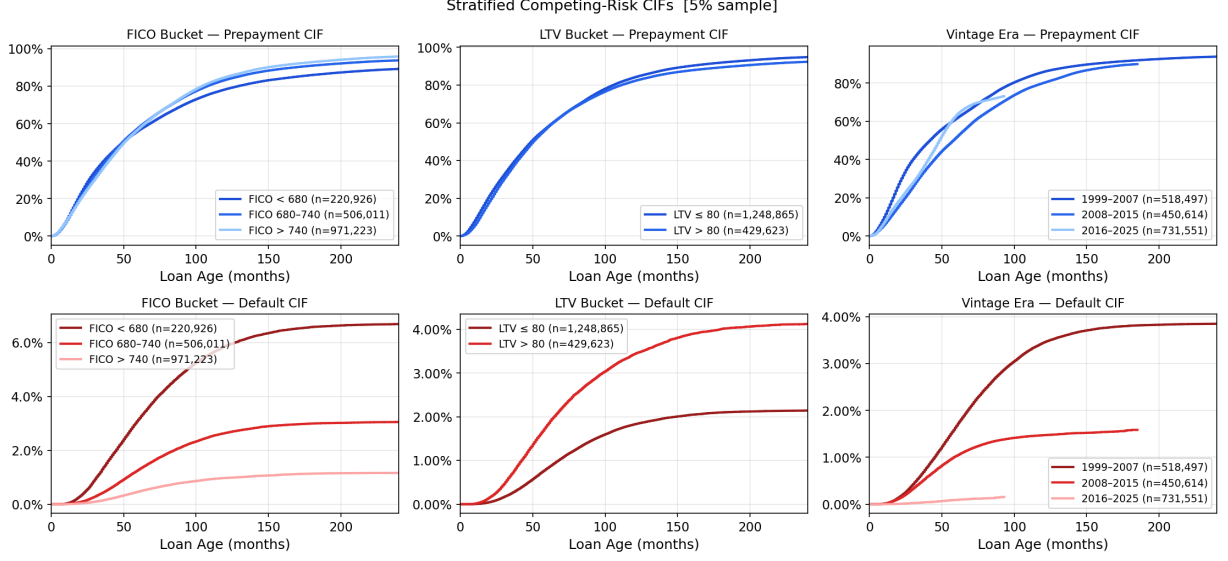
**LTV** doubles default risk ( $>80$ :  $\approx 4.5\%$  vs  $\approx 2\%$  at 20 yr) while leaving prepayment curves nearly identical. LTV blocks the refinancing path (negative equity) but not exit by sale or payoff.

**Vintage era** is the dominant stratifier. GFC vintages (1999–2007) reach  $\approx 4\%$  default CIF; post-2013 approaches zero. Post-2020 prepayment CIF is nearly flat: rate lock-in.

## 6. Cause-Specific Cox Regression

Two independent Cox models fit one per cause, treating the alternative as censored:

$$h_k(t \mid \mathbf{x}) = h_{0k}(t) \exp(\beta_k^\top \mathbf{x}), \quad k \in \{\text{pre}, \text{def}\}. \quad (3)$$



**Figure 5:** Aalen–Johansen CIF by FICO (top), LTV (middle), and vintage era (bottom). Left: prepayment. Right: default.

Partial likelihood:

$$\ell(\beta_k) = \sum_{i: E_i=k} \left[ \beta_k^\top \mathbf{x}_i - \log \sum_{j \in \mathcal{R}(t_i)} e^{\beta_k^\top \mathbf{x}_j} \right]. \quad (4)$$

14 features: 10 numeric (credit score, LTV, DTI, UPB, origination-era macro, rate incentive, max delinquency, ever 90+ DQ) + 4 one-hot (loan purpose  $\times 2$ , occupancy  $\times 2$ );  $\ell_2$  penalizer = 0.1.

| Covariate       | Pre $\hat{\beta}$ | Def $\hat{\beta}$ | Interpretation                                        |
|-----------------|-------------------|-------------------|-------------------------------------------------------|
| Rate incentive  | +0.256            | +0.021            | Refinancing driver; no default content                |
| Mortgage rate   | +0.250            | +0.060            | High orig. rate $\rightarrow$ future refi opportunity |
| UPB             | +0.161            | −0.010            | Fixed refi cost amortized over larger balance         |
| FICO            | +0.010            | −0.047            | <b>Opposite:</b> access to refi vs. credit quality    |
| LTV             | −0.011            | +0.041            | <b>Opposite:</b> equity enables refi; traps defaulter |
| Max Delinquency | −0.245            | +0.298            | <b>Opposite:</b> distressed loans default, skip refi  |
| Inv. property   | −0.297            | +0.270            | <b>Opposite:</b> investors walk rather than refi      |

The *rate incentive* coefficient ( $e^{0.256} \approx 1.29$ ) captures the classic prepayment S-curve: a 1-SD increase in the rate spread raises prepayment hazard by 29%. Its near-zero default coefficient (+0.021) confirms it carries no credit-risk information.

## 7. Fine–Gray Subdistribution Hazard

Fine–Gray directly models the CIF by retaining already-defaulted loans in the risk set with IPCW weights  $w_i(t) = \hat{G}(t_i)/\hat{G}(\min(t_i, t))$ :

$$\hat{F}_k(t) = 1 - \exp\left(-\int_0^t \tilde{h}_{0k}(s) e^{\gamma_k^\top \mathbf{x}} ds\right). \quad (5)$$

The maximum coefficient difference vs. cause-specific Cox is  $|\Delta| = 0.004$ . With only 1.6% default rate, the IPCW-weighted competing-event borrowers contribute negligibly:  $\tilde{h}_1(t) \approx h_1(t)$  when  $F_2(t) \rightarrow 0$ .

**Use case:** cause-specific Cox for causal mechanism estimation; Fine–Gray for CIF prediction and MBS loss forecasting.

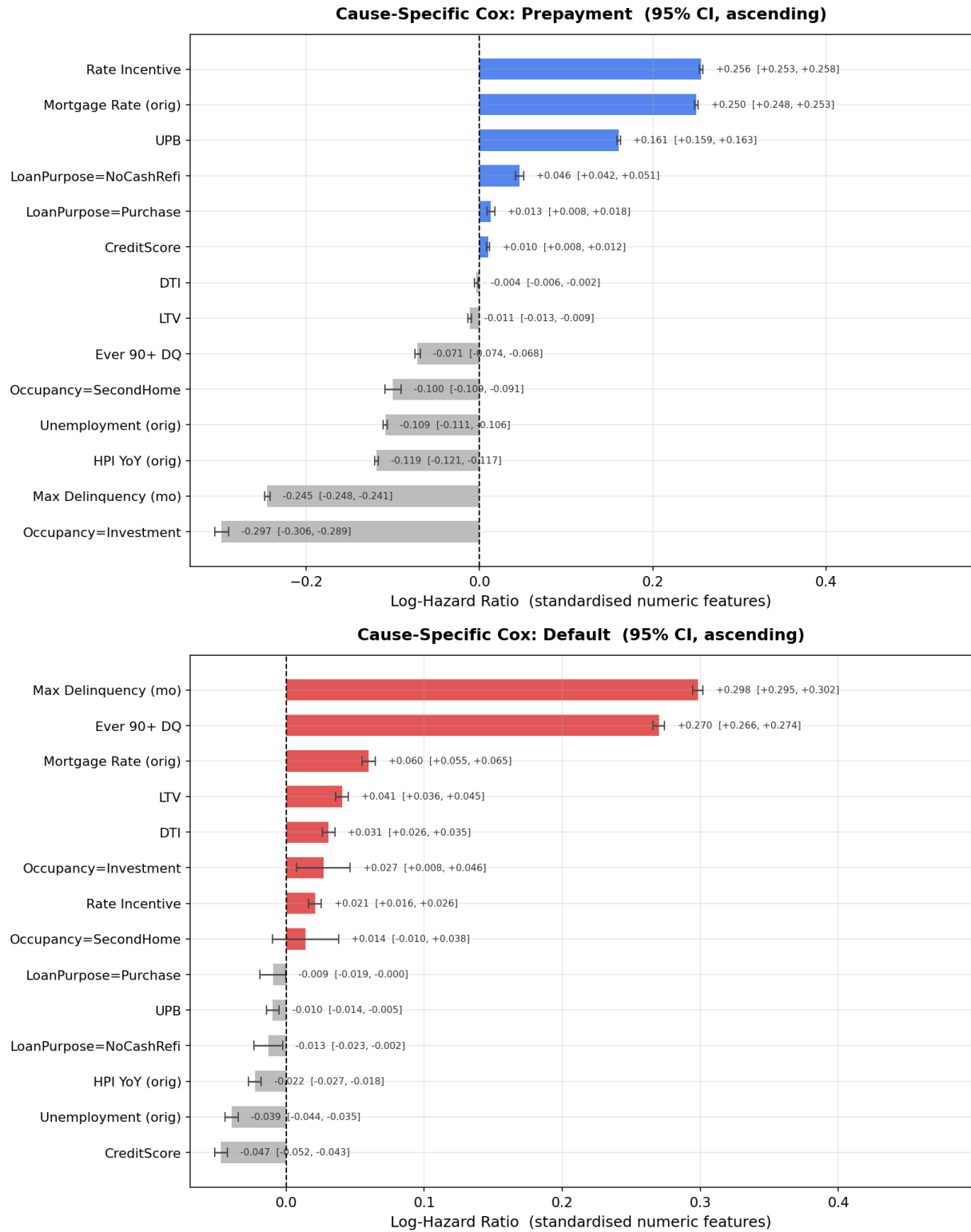
## 8. Key Findings and Insights

1. **Competing risks are not optional.** Four covariates (FICO, LTV, Max Delinquency, Investment property) have opposite signs between prepayment and default. A combined model cancels both signals.
2. **KM bias reaches 2.2 pp at 20 years.** Systematically over-stating the prepayment probability misprices the embedded duration option in MBS —  $\approx \$1\text{M}$  per  $\$100\text{M}$  pool at the 10-year horizon.
3. **Vintage era dominates all other risk stratifiers.** GFC-era (1999–2007) default CIF  $\approx 4\%$  vs  $\approx 0$  post-2013. Post-2020 prepayment CIF is flat: rate lock-in has suspended the refi cycle.
4. **FICO separates default  $6\times$ ; it barely separates prepayment.** FICO belongs in a default model. A high-FICO borrower still needs a rate incentive and sufficient equity to actually refinance.
5. **Rate incentive is the dominant, cause-pure prepayment signal.**  $\hat{\beta} = +0.256$  ( $e^{0.256} \approx 1.29$ ) for prepayment;  $\approx 0$  for default. All rate-shock scenario analysis should route through this coefficient.
6. **Fine–Gray  $\approx$  cause-specific Cox in this dataset; the distinction matters in crisis pools.** With 1.6% default, IPCW weights are near 1. In subprime or GFC-era pools with high competing-event rates, Fine–Gray would give materially different CIF forecasts.

**Priority improvements:** (1) Replace static “Max Delinquency” with a rolling 3-month delinquency indicator. (2) Add an  $\text{LTV} \times \text{rate\_incentive}$  interaction: high-LTV borrowers cannot refi even when incentivized. (3) Stratify  $h_{0k}(t)$  by vintage era to capture cohort-specific burnout patterns.

## References

- [1] Fine, J. P. and Gray, R. J. (1999). *JASA* 94:496–509.
- [2] Aalen, O. O. and Johansen, S. (1978). *Scand. J. Statistics* 5:141–150.
- [3] Sadhwani, A., Giesecke, K. and Sirignano, J. (2021). *J. Financial Econometrics* 19:313–368.



**Figure 6:** Cause-specific Cox forest plots. Top: prepayment model. Bottom: default model. Note the sign reversals across the two panels.